

Understanding Retrieval Augmented Generation

Assignment Report — AI Agent Engineer Roadmap | edQuest | June 2026

1. Introduction

RAG stands for Retrieval Augmented Generation. It is a method used in AI systems where the model does not just rely on what it learned during training, but also fetches relevant information from an external source before giving an answer. Regular AI models have a knowledge cutoff and sometimes make up wrong answers. RAG solves both problems by searching for real information first and then generating the answer based on that.

2. Why RAG is Important

- It lets the AI work with private or company-specific data that was not part of its training.
- It reduces wrong or made-up answers by grounding responses in real retrieved documents.
- It is cost-effective — only the knowledge base needs updating, not the whole model.
- It can cite the source of its answer, which builds trust in professional settings.

3. How RAG Works

Step 1 — Indexing: Documents are broken into smaller chunks, converted into embeddings (numbers representing meaning), and stored in a vector database.

Step 2 — Retrieval: When a user asks a question, it is also converted to an embedding. The system searches the database and finds the most relevant chunks.

Step 3 — Generation: The retrieved chunks are passed to the AI along with the question. The AI then generates an answer based on the actual information found, rather than guessing.

4. Data Retrieval Techniques

Vector Search (Dense Retrieval): Converts both the question and documents into vectors and finds the closest match in meaning. It is good at understanding intent, not just exact words.

Keyword Search (Sparse Retrieval): Methods like BM25 look for exact word matches. Faster but less accurate than vector search for natural language.

Hybrid Search: Most real-world systems combine both methods to get better overall accuracy and precision.

5. Applications of RAG

- **Customer Support:** Chatbots retrieve answers from help documents instead of guessing.
- **Healthcare:** Doctors can query medical databases and get answers grounded in clinical guidelines.

- **Legal:** Law firms search case files and regulations before drafting documents.
- **Enterprise:** Employees can search internal company knowledge bases using plain language.

6. Case Studies

Shopify — Customer Support: Shopify built a RAG-based AI assistant that searches through their help documentation. When a merchant asks something, the system retrieves the most relevant help articles and gives an accurate, specific answer. This helped Shopify handle more queries without increasing support staff.

Morgan Stanley — Financial Research: Morgan Stanley built an AI assistant that searches over 100,000 internal research reports and financial documents. Financial advisors get answers based on real company research before client meetings, saving time and ensuring accuracy.

Notion — Personal Workspace AI: Notion's AI reads the user's own notes, task lists, and goals before generating content. Instead of a generic response, it gives something that is relevant to what the user has already written in their workspace.

7. My Proposed Application — RAG for Exam Preparation

My idea is to use RAG for personalized exam coaching. The system would index two things — the full exam syllabus with study material, and the student's own answer history and mistake records. When a student gets a question wrong, the system retrieves the exact topic explanation from the study material and also checks what similar mistakes that student made before. The AI then gives a short, focused explanation based on their specific weak area.

This is different from existing tools because it retrieves from the student's own performance data, not just general content. Most AI tutors give the same explanation to everyone. This would feel like a personal tutor who actually knows your weak points. It would be especially useful for competitive exams like GATE, UPSC, or medical entrance tests.

8. Conclusion

RAG is a simple but very powerful idea — search for the right information first, then generate the answer using that. This makes AI more accurate, grounded, and trustworthy. The case studies show that companies are already getting real value from RAG across customer support, finance, and productivity. There is still a lot of room to grow, especially in education. I believe RAG-based systems will become a standard part of most AI applications in the coming years.

References

1. Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Meta AI.
2. Gao, Y., et al. (2023). RAG for Large Language Models: A Survey. arXiv:2312.10997.